

SEQUENCES, SERIES AND DIFFERENTIAL EQUATIONS

Gilberto Berardinelli

Department of Electronic Systems, Aalborg University, Denmark



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Lecture format



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- Lecture (approx. 1h45m including break) + exercise session
- Slides + large usage of blackboard
- No pre-scheduled breaks
 - Breaks are "distributed" according to the complexity of the presented topics
- Please let me know if I erase the blackboard too quickly!

Plan for the course

- Sequences and series
 - Convergence tests
- Power series
 - Convergence radius
- Taylor and McLaurin series
- Differential equations
 - Ordinary differential equations
 - First and second order differential equations
 - Using Laplace transform for solving differential equations

Literature

Erwin Kreyszig
ADVANCED ENGINEERING MATHEMATICS, 10th edition
Wiley, 2011

Robert A. Adams and Christopher Essex,
"Calculus; a complete course", Pearson, 9th edition,
2010.



Today's agenda

- Sequences
- Series
- Convergence tests
 - Comparison test
 - Ratio test
 - Root test
 - Integral test
- Exercises!

Sequences

- A sequence is obtained by assigning to each positive integer n a number z_n .

e.g., $n=1,2,3,4,5 \rightarrow z_1=4.3, z_2=15.2, z_3=23, z_4=31.7, z_5=55.4$

- A real sequence is a sequence where all the terms are real.
- A complex sequence is a sequence whose terms are complex.

Sequences - Convergence



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Convergence. A convergent sequence $z_1, z_2, \dots$ is one that has a limit c , written

$$\lim_{n \rightarrow \infty} z_n = c \quad \text{or simply} \quad z_n \rightarrow c.$$

By definition of **limit** this means that for every $\epsilon > 0$ we can find an N such that

$$(1) \quad |z_n - c| < \epsilon \quad \text{for all } n > N;$$

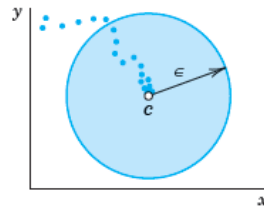


Fig. 361. Convergent complex sequence

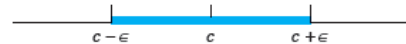


Fig. 362. Convergent real sequence

Sequences of the Real and the Imaginary Parts

A sequence $z_1, z_2, \dots, z_n, \dots$ of complex numbers $z_n = x_n + iy_n$ (where $n = 1, 2, \dots$) converges to $c = a + ib$ if and only if the sequence of the real parts $x_1, x_2, \dots$ converges to a and the sequence of the imaginary parts $y_1, y_2, \dots$ converges to b .

Series

- An (ordered) infinite sequence ($z_1, z_2, z_3, \dots$) of terms (that is, numbers, functions) defines a **series**, which is the operation of adding the z_i terms one after the other.

$$\sum_{m=1}^{\infty} z_m = z_1 + z_2 + \dots$$

- Partial sums can be

$$s_1 = z_1, \quad s_2 = z_1 + z_2, \quad s_3 = z_1 + z_2 + z_3, \quad \longrightarrow \quad s_n = z_1 + z_2 + \dots + z_n \quad (n = 1, 2, \dots).$$

- A **convergent series** is the one whose sequence of partial sums converges.

$$\lim_{n \rightarrow \infty} s_n = s. \quad \longrightarrow \quad s = \sum_{m=1}^{\infty} z_m = z_1 + z_2 + \dots$$

Series – Convergence and divergence

Divergence

If a series $z_1 + z_2 + \dots$ converges, then $\lim_{m \rightarrow \infty} z_m = 0$. Hence if this does not hold, the series diverges.

Necessary but not sufficient condition!

i.e. there are cases where $\lim_{m \rightarrow \infty} z_m = 0$, but the series diverges!

Series - Convergence

Absolute Convergence. A series $z_1 + z_2 + \cdots$ is called **absolutely convergent** if the series of the absolute values of the terms

$$\sum_{m=1}^{\infty} |z_m| = |z_1| + |z_2| + \cdots$$

is convergent.

If $z_1 + z_2 + \cdots$ converges but $|z_1| + |z_2| + \cdots$ diverges, then the series $z_1 + z_2 + \cdots$ is called, more precisely, **conditionally convergent**.

Absolute convergence $\rightarrow$ convergence



Today's agenda

- Sequences
- Series
- **Convergence tests**
 - Comparison test
 - Ratio test
 - Root test
 - Integral test
- Exercises!

Series - Convergence test



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Comparison Test

If a series $z_1 + z_2 + \dots$ is given and we can find a convergent series $b_1 + b_2 + \dots$ with nonnegative real terms such that $|z_1| \leq b_1$, $|z_2| \leq b_2, \dots$, then the given series converges, even absolutely.

Series – Ratio test



Ratio Test

If a series $z_1 + z_2 + \dots$ with $z_n \neq 0$ ($n = 1, 2, \dots$) has the property that for every n greater than some N ,

$$(7) \qquad \left| \frac{z_{n+1}}{z_n} \right| \leq q < 1 \qquad (n > N)$$

(where $q < 1$ is fixed), this series converges absolutely. If for every $n > N$,

$$(8) \qquad \left| \frac{z_{n+1}}{z_n} \right| \geq 1 \qquad (n > N),$$

the series diverges.

Ratio Test

If a series $z_1 + z_2 + \dots$ with $z_n \neq 0$ ($n = 1, 2, \dots$) is such that $\lim_{n \rightarrow \infty} \left| \frac{z_{n+1}}{z_n} \right| = L$, then:

- (a) If $L < 1$, the series converges absolutely.
- (b) If $L > 1$, the series diverges.
- (c) If $L = 1$, the series may converge or diverge, so that the test fails and permits no conclusion.

Series – Root test

Root Test

If a series $z_1 + z_2 + \cdots$ is such that for every n greater than some N ,

$$(9) \quad \sqrt[n]{|z_n|} \leq q < 1 \quad (n > N)$$

(where $q < 1$ is fixed), this series converges absolutely. If for infinitely many n ,

$$(10) \quad \sqrt[n]{|z_n|} \geq 1,$$

the series diverges.

Root Test

If a series $z_1 + z_2 + \cdots$ is such that $\lim_{n \rightarrow \infty} \sqrt[n]{|z_n|} = L$, then:

- (a) The series converges absolutely if $L < 1$.
- (b) The series diverges if $L > 1$.
- (c) If $L = 1$, the test fails; that is, no conclusion is possible.

Series – Integral test

- Given a function f continuous, positive, decreasing in $[0, \infty)$ such that $z_n = f(n)$

$\sum_{n=N}^{\infty} z_n$ converges if and only if $\int_N^{\infty} f(n) \, dn$ converges.

Series – Alternating series test

- A series of the following form:

$$\sum_{n=0}^{\infty} (-1)^n a_n$$

where the a_n coefficients are either all positive or all negative, is an alternating series.

If the following conditions hold:

- 1) $|a_n|$ coefficient decrease monotonically with n
- 2) $\lim_{n \rightarrow \infty} a_n = 0$

Then the series converges.

Note however, that the series might not converge absolutely, but only conditionally!

Series



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- An important series!

Geometric Series

The geometric series

(6*)
$$\sum_{m=0}^{\infty} q^m = 1 + q + q^2 + \cdots$$

converges with the sum $1/(1 - q)$ if $|q| < 1$ and diverges if $|q| \geq 1$.